

Northeastern University
Mechanical and Industrial Engineering Department
IE 6200: Engineering Probability and Statistics
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Impact of Key Factors on Team Success in India's Cricket Sport League - IPL

Project Progress Report



Northeastern University

Group 4

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Problem Statement

Cricket is one of the world's most popular sports, featuring two 11-player teams. One team bats to score runs, while the other bowls and fields to restrict runs and take wickets. Each team has a certain amount of overs to bat and bowl. The team that scores the most runs or effectively prevents their opponent from reaching the target wins the match.

The Indian Premier League (IPL), a franchise-based Twenty20 (Twenty-Twenty) league, debuted in 2008 and has since transformed the sport by incorporating worldwide talent, high-stakes auctions, and creative match strategy. The T20 format allows each team to play a maximum of 20 overs, making the game fast-paced, exciting, and competitive. While certain teams continuously dominate the competition, others struggle to sustain consistent performance, raising the question of what genuinely drives a team's success or failure in the IPL.

This research tries to reveal an IPL team's performance trends across multiple seasons by analyzing match data and identifying patterns that contribute to victory or defeat.

Project Goals

The objective of this project is to analyze historical data from the Indian Premier League (IPL) seasons spanning 2020 to 2024 to uncover key factors that influence a team's success or failure. The project aims to collect comprehensive match data, including team performance metrics, match conditions, individual statistics, and investment aspects. Statistical techniques will be applied to conduct both descriptive and inferential analyses.

1. Comprehensive Evaluation of Team Performance:

The analysis will assess CSK's performance across various seasons using statistical methods like mean, median, standard deviation, and range. These metrics will provide insights into key indicators such as runs scored, wickets taken, and win percentages, helping to evaluate the team's consistency and performance under different match conditions.

2. Identification of Performance Determinants:

The study will focus on identifying factors that influence match outcomes. Key metrics such as batting and bowling averages, toss results, and match venues will be analyzed to determine their impact on CSK's success or failure, enabling more informed strategic decisions.

3. Trend Analysis:

Venue-based trend analysis will be a critical part of the study, focusing on stadiums located across various Indian states where CSK has played frequently, enabling them to identify location-specific scoring strategies for better match outcomes.

4. Analysis of Factors Affecting Match Results:

This section evaluates key elements influencing CSK's match outcomes, including batting, bowling, toss results, venue performance, and home vs. away success using statistical methods.

Data Collection and Attributes

Data Sources

The dataset for this research was compiled from several publicly available sources, including Kaggle ([Kaggle IPL Dataset – All Seasons](#)), Howstat ([Howstat IPL Match List](#)) and IPL ([Official Website](#)) It compiles match data for the Chennai Super Kings, an Indian Premier League (IPL) team, across five seasons (2020-2024). This dataset contains precise information about team performance, match results, and essential performance measures.

Match_ID	Season (Year)	Opponent Team	Venue	Toss_Winner	Toss_Decision	Match_Winner	Winner Team (Own/O)	Match_Win_Ty
1216492	Season-2020	Mumbai Indians	Sheikh Zayed Stadium	1	2	Chennai Super Kings	1	wickets
1216506	Season-2020	Kings XI Punjab	Sheikh Zayed Stadium	1	2	Chennai Super Kings	1	wickets
1216513	Season-2020	Kings XI Punjab	Dubai International Cricket Stadium	2	1	Chennai Super Kings	1	wickets
1216528	Season-2020	Sunrisers Hyderabad	Dubai International Cricket Stadium	1	1	Chennai Super Kings	1	runs
1216536	Season-2020	Kolkata Knight Riders	Dubai International Cricket Stadium	1	2	Chennai Super Kings	1	wickets
1216544	Season-2020	Royal Challengers Bangalore	Dubai International Cricket Stadium	2	1	Chennai Super Kings	1	wickets
1254065	Season-2021	Punjab Kings	Wankhede Stadium, Mumbai	1	2	Chennai Super Kings	1	wickets
1254069	Season-2021	Rajasthan Royals	Wankhede Stadium, Mumbai	2	2	Chennai Super Kings	1	runs
1254072	Season-2021	Kolkata Knight Riders	Wankhede Stadium, Mumbai	2	2	Chennai Super Kings	1	runs
1254076	Season-2021	Royal Challengers Bangalore	Wankhede Stadium, Mumbai	1	1	Chennai Super Kings	1	runs
1254080	Season-2021	Sunrisers Hyderabad	Arun Jaitley Stadium, Delhi	2	1	Chennai Super Kings	1	wickets
1254091	Season-2021	Sunrisers Hyderabad	Sharjah Cricket Stadium	1	2	Chennai Super Kings	1	wickets
1254098	Season-2021	Kolkata Knight Riders	Zayed Cricket Stadium, Abu Dhabi	2	1	Chennai Super Kings	1	wickets
1254104	Season-2021	Mumbai Indians	Dubai International Cricket Stadium	1	1	Chennai Super Kings	1	runs
1254113	Season-2021	Royal Challengers Bangalore	Sharjah Cricket Stadium	1	2	Chennai Super Kings	1	wickets
1254114	Season-2021	Delhi Capitals	Dubai International Cricket Stadium	1	2	Chennai Super Kings	1	wickets
1254117	Season-2021	Kolkata Knight Riders	Dubai International Cricket Stadium	2	2	Chennai Super Kings	1	runs
1304068	Season-2022	Royal Challengers Bangalore	Dr DY Patil Sports Academy, Mumbai	2	2	Chennai Super Kings	1	runs
1304079	Season-2022	Mumbai Indians	Dr DY Patil Sports Academy, Mumbai	1	2	Chennai Super Kings	1	wickets
1304092	Season-2022	Sunrisers Hyderabad	Maharashtra Cricket Association Stadium, Pune	2	2	Chennai Super Kings	1	runs
1304101	Season-2022	Delhi Capitals	Dr DY Patil Sports Academy, Mumbai	2	2	Chennai Super Kings	1	runs
1359480	Season-2023	Lucknow Super Giants	MA Chidambaram Stadium, Chepauk, Chennai	2	2	Chennai Super Kings	1	runs
1359486	Season-2023	Mumbai Indians	Wankhede Stadium, Mumbai	1	2	Chennai Super Kings	1	wickets
1359498	Season-2023	Royal Challengers Bangalore	M Chinnaswamy Stadium, Bengaluru	2	2	Chennai Super Kings	1	runs
1359503	Season-2023	Sunrisers Hyderabad	MA Chidambaram Stadium, Chepauk, Chennai	1	2	Chennai Super Kings	1	wickets
1359507	Season-2023	Kolkata Knight Riders	Eden Gardens, Kolkata	2	2	Chennai Super Kings	1	runs
1359519	Season-2023	Lucknow Super Giants	Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cricket Stadium, Lucknow	1	2	Draw/No Result	2	

Fig0: Collected Dataset of CSK Team

The dataset shown in figure 0 includes detailed match-level data for Chennai Super Kings (CSK) across five IPL seasons (2020-2024). It consists of multiple subsheets: Master Sheet (aggregated data from all seasons), Season-2020 to Season-2024 (season-specific match

data), and sheets highlighting Win by Runs and Win by Wickets, showing how CSK secured victories. Additionally, the IPL Performance sheet provides an overview of CSK's league stage positions, win/loss records, and title wins over the seasons. Key variables across all sheets include match details, toss outcomes, match winner, and win type (runs or wickets), enabling comprehensive analysis of CSK's performance, strategies, and trends across seasons.

Data Source and Selection Process

- **Multi-Source Data Compilation:** To guarantee a full and accurate depiction of match statistics, the dataset was constructed by gathering and combining data from numerous cricket analytics sources, including Kaggle, the official IPL website, and Howstat.
- **Consolidated Dataset Approach:** Instead of using numerous independent datasets, the data has been merged into a single master dataset for more efficient analysis.
- **Automated and Verified Data Extraction:** The dataset was acquired from known cricket statistics platforms and formatted systematically.

Data Cleaning and Processing

The dataset is organized in a table manner, with rows representing individual matches and columns containing match-specific statistics. The data has been refined to ensure correctness and consistency:

- **Match identifiers:** Match identifiers are standardized naming conventions for teams and venues.
- **Performance metrics:** Performance measures include averages of first and second innings scores.
- **Missing values:** Missing values were removed or interpolated if appropriate.
- **Data formatting:** Ensured that numerical values and category variables were consistent.

Assumptions Considered

1. **Match Completeness:** All match data is presumed to be complete and accurate based on official records.
2. **Matches are given equal weight:** Each match is seen as equally important, regardless of opponent strength.
3. **Exclusion of Economy and Strike Rates:** Because the dataset lacks precise bowling economy rates and batting strike rates, they are not included in this study.

4. Consistent Weather Conditions: The majority of IPL matches take place during the summer months, and weather conditions stay rather consistent among seasons.
5. Accurate Data Sources: Because the dataset was generated from well-known cricket analytics tools, it is deemed to be reliable.

Key Variables in the Dataset

1. Independent Variable:
 - Season: The year the match was played (2020-2024).
2. Dependent Variables:
 - Second Innings Score: The total number of runs scored by the side batting second.
 - Powerplay scores are runs scored in the first six overs of an innings.
 - Middle Overs Scores: Runs scored between overs 7 and 15.
 - Death Overs Scores: Runs scored in the last five overs (16-20).
 - Win Margin: The number of runs or wickets used to win the match.
 - First Innings Score: The number of runs scored by the team batting first.
3. Categorical Variables:
 - Opponent Team: The team that the Chennai Super Kings faced.
 - Venue: The stadium where the game took place.
 - Toss Winner: The team who won the toss.
 - Toss Decision: The team that wins the toss decides whether to bat or bowl.
 - contest Winner: The team who won the contest.
 - Win Type: Whether the match was won with runs or wickets.

Data Refinement and Standardization

1. Incomplete Data Removal: Matches with missing statistics were removed to maintain data accuracy.
2. Match Outcome Classification: A standardized win-loss classification based on match margins.
3. Home vs Away Differentiation: Not expressly present in the dataset, but inferred from venues.
4. Numerical Standardization: All numerical values (scores and margins) were formatted similarly.

Data Visualization

Horizontal Bar Chart - IPL Performance Landscape

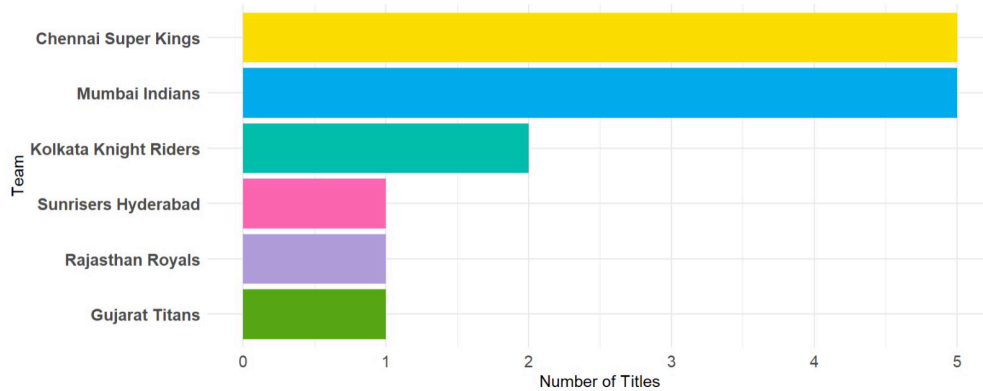


Fig1: IPL Trophies won by teams from 2008 to 2024

Figure 1 is a horizontal bar chart showing the number of IPL trophies won by teams from 2008 to 2024. Chennai Super Kings and Mumbai Indians lead with 5 titles each, highlighting their dominance in the league. Kolkata Knight Riders follow with 2 titles, while Sunrisers Hyderabad, Rajasthan Royals, and Gujarat Titans have won 1 title each. The chart provides a clear comparison of team success over the years.

Grouped Bar Chart - Seasonal Performance Trends

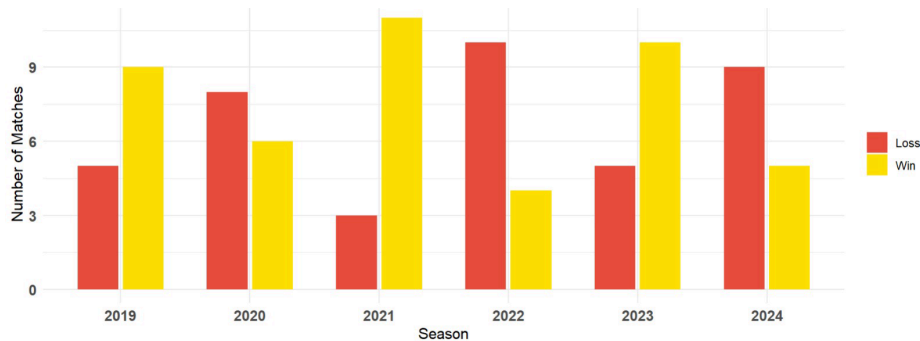


Fig2: CSK Wins and Losses last 5 seasons from 2020 to 2024

Figure 2 is a grouped bar chart illustrating CSK's seasonal performance in terms of wins and losses over the last five IPL seasons (2020 to 2024). Each season shows two bars—one for wins (in yellow) and one for losses (in red). The chart reveals fluctuations in performance, with standout success in 2021 and 2023, while 2022 marked the lowest win count. This visual effectively highlights trends in CSK's consistency and competitiveness across seasons.

Mosaic Plot - Toss vs Match Outcome

Analysis of the relationship between toss result and match outcome for CSK using a mosaic plot shown in figure 3. Of the 76 total matches analyzed, CSK won 18 matches after winning the toss and 21 matches after losing it. Similarly, they lost 15 matches after

winning the toss and 22 after losing it. These frequencies suggest no strong directional advantage from winning the toss.

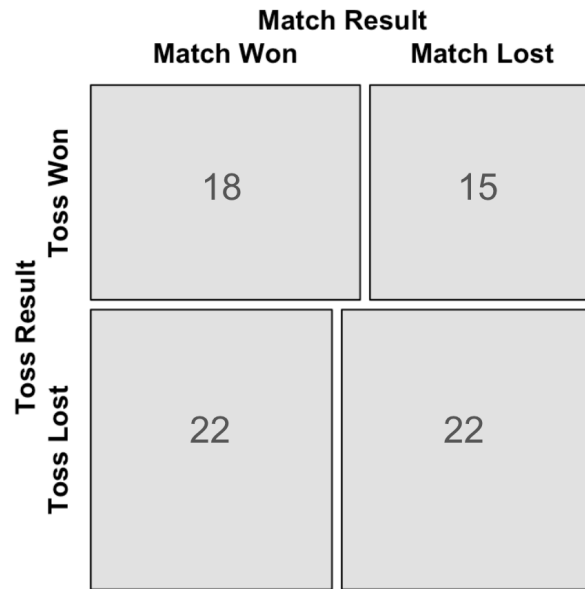


Fig3: Mosaic Plot of CSK Toss and Match Wins

This insight is visually represented in the mosaic plot, where the segment sizes appear proportionally distributed, indicating a weak or no dependency between toss and match result.

Box Plot - Powerplay Scores

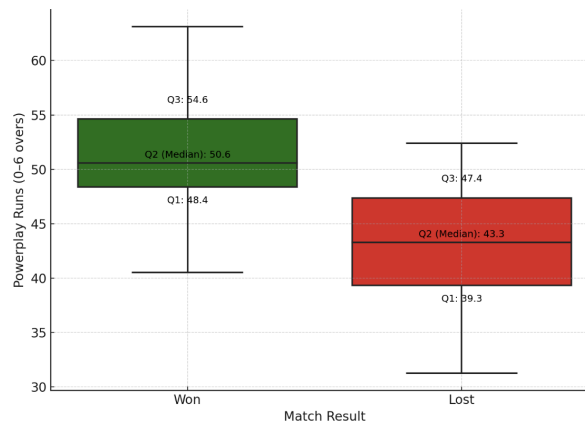


Fig4: Boxplot comparing powerplay scores in CSK wins vs losses

Figure 4 presents a boxplot comparing the Powerplay runs (0-6 overs) in matches won and lost by Chennai Super Kings (CSK) in the IPL. The chart shows that in matches won, CSK typically scores higher during the Powerplay, with a median score of 50.6 runs. The scores

are relatively consistent, as indicated by a narrow interquartile range (48.4 to 54.6). In contrast, during matches lost, CSK's Powerplay performance is weaker, with a lower median score of 43.3 runs and a broader range (39.3 to 47.4). This suggests that a strong and consistent Powerplay performance may be a key factor influencing CSK's success in IPL matches.

Box Plot - CSK's First Innings Score Comparison

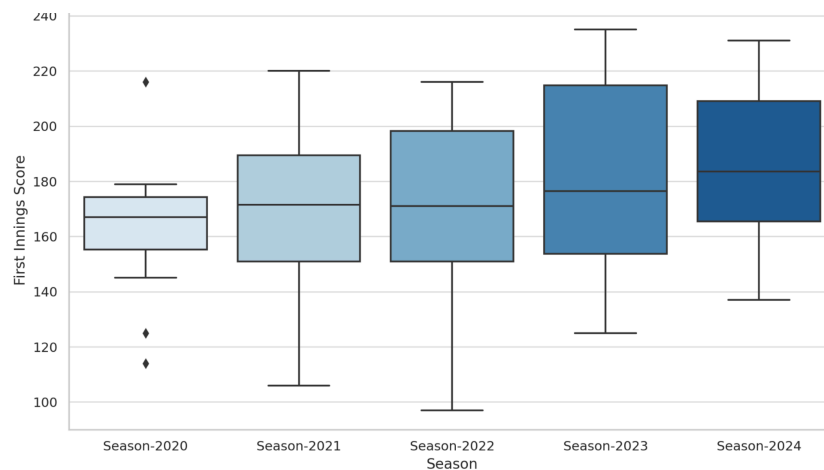


Fig5: Comparison of CSK's first innings scores over different seasons (2020-2024)

The box plot illustrates CSK's first innings scores across IPL seasons 2020 to 2024 shown in figure 5. A noticeable upward trend is observed from 2020 to 2023, with 2023 showing the highest variability and peak scores. While 2020 had a lower and more consistent scoring pattern, later seasons, especially 2023 and 2024, reflect stronger and more aggressive batting performances. This trend suggests CSK's evolving strategy and improved batting depth over the years.

Scatter Plot - Correlation between CSK powerplay and first innings scores

To understand how early momentum in the match influences the overall batting outcome, we analyzed the relationship between Powerplay scores (first 6 overs) and total First Innings scores for all CSK matches over the last five IPL seasons. The scatterplot shown in figure 6 with a fitted regression line reveals a moderate positive correlation ($r = 0.46$), suggesting that higher scores in the Powerplay phase tend to be associated with higher total scores in the innings. The coefficient of determination ($r^2 = 0.21$) indicates that approximately 21% of the variability in CSK's first innings total can be explained by Powerplay performance alone. While not perfectly predictive, this relationship supports the idea that a strong start contributes significantly to building a competitive total, offering valuable insights for CSK's batting strategy.

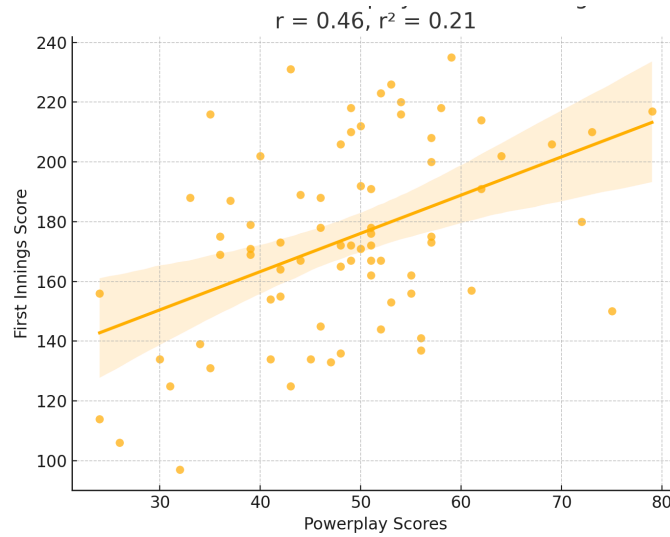


Fig6: Correlation between CSK powerplay and first innings scores

Statistical Analysis

Sampling Distribution

a. Sampling Distribution of Means - Powerplay Scores

The sampling distribution of the mean Powerplay scores (first six overs) for CSK based on 1000 samples, each of size 30, is depicted in the histogram shown in figure 7. The distribution clusters around a core value close to 48 and is roughly symmetrical and bell-shaped.

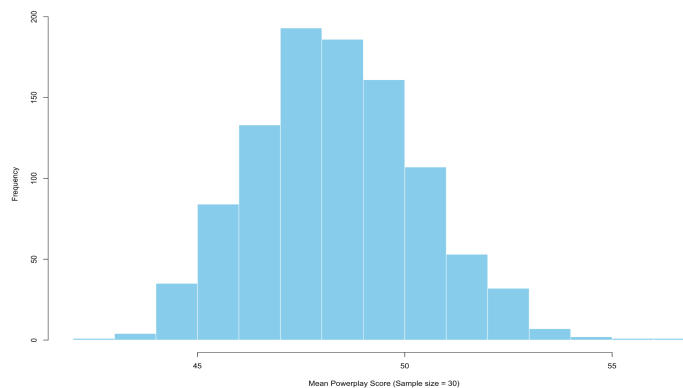


Fig 7: Sampling Distribution of Means - Powerplay Scores

The Central Limit Theorem (CLT) is validated by this graph. The means of sufficiently large random samples create a roughly normal distribution, even though individual match scores may be skewed. The majority of data fall between 46 and 50, indicating that CSK usually performs in this area during Powerplay. Relatively stable early-innings performance with fewer extreme outliers is implied by the steady peak around the mean. When assessing CSK's strategic aggression and consistency throughout the Powerplay phase, this information is helpful.

b. Sampling Distribution of Variance - DeathOver scores

Based on 1000 samples of size 30, this histogram shown in figure 8 displays the distribution of sample variations in CSK's death-over scores (overs 16–20). With a lengthy tail extending toward higher variance values, the shape is right-skewed.

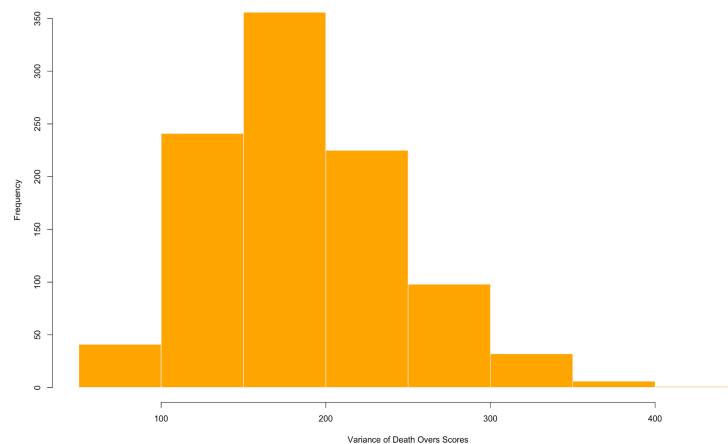


Fig 8: Sampling Distribution of Variance - DeathOver scores

While many samples have moderate variability, the skewed nature shows that some have relatively high variation, indicating inconsistent or explosive performances toward the end. The most frequent degree of score volatility is indicated by the peak around variance = 175–200. This implies that while CSK's closing phase is usually successful, it can vary significantly based on the circumstances of the game or the quality of the opposition. Matches with particularly noteworthy individual death-over contributions may be represented by the right tail.

c. Sampling Distribution of Proportion - CSK wins when Batting First

The distribution of the percentage of CSK victories when the team decides to bat first is shown in the plot of figure 9. The percentage of wins in one of 1000 random samples (size 30) is represented by each point. With the majority of values falling between 0.10 and 0.25, the form is slightly tilted to the right.

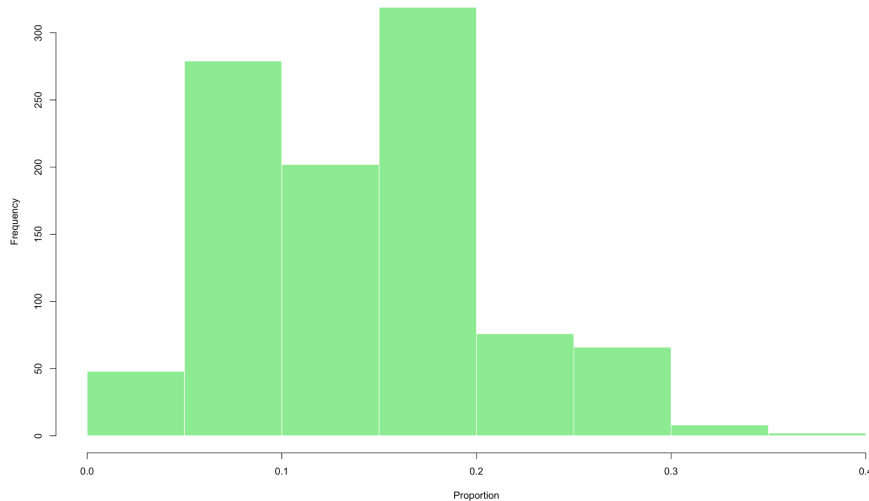


Fig 9: Sampling Distribution of Proportion - CSK wins when Batting First

According to the data, CSK often has a success rate of less than 25% while batting first in the games that were sampled, indicating a predilection or strength in chasing targets. When establishing a target, the low concentration of samples with large proportions indicates restricted dominance. The asymmetry supports occasional successful first-innings defenses, but not as a persistent tendency. Particularly on batting-friendly pitches, this knowledge could help with choices like selecting to field first after winning the toss.

d. Sampling Distribution of Proportion- When Batting Second Or Bowling First:

The histogram shown in figure 10, which was produced using 1000 random samples with a sample size of 30, shows the sampling distribution of the Chennai Super Kings' (CSK) percentage of victories when batting second. The majority of sample proportions fall between 0.25 and 0.45, indicating a strongly right-skewed distribution. The most common win % in the sampling process is indicated by the peak, which falls between 0.35 and 0.40. The distribution's form indicates a steady chance of winning when pursuing.

Statistically speaking, CSK has a higher victory record when chasing (batting second) than when setting a target. A higher chance of winning when chasing is indicated by a proportional concentration of about 35–40%. A slight skew to the right indicates that some matches had outstanding finishes when pursuing targets. This reinforces CSK's established reputation as a squad that performs well under duress, especially when clever finishers lead successful run chases.

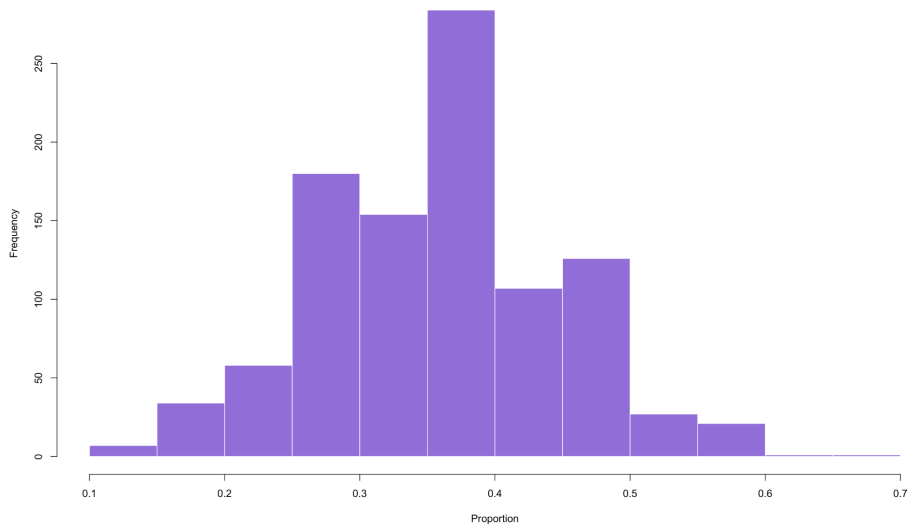


Fig 10: Sampling Distribution of Proportion

According to the sample distribution, CSK's success rate when batting second is significantly greater than when batting first. This points to a tactical edge in chasing, which is consistent with past assessments of the team's performance under duress, especially in seasons when seasoned finishers like MS Dhoni and Ravindra Jadeja were in charge (Star Performers of the Franchise). This study suggests that choosing to field first after winning the toss increases the chances of CSK prevailing.

Confidence Interval:

1. Confidence Interval of Mean

a. To estimate the average Powerplay score (first 6 overs) that Chennai Super Kings (CSK) achieved in matches they won over the last five IPL seasons.

N	Mean	StDev	SE Mean	95% CI for μ
38	49.05	10.38	1.68	(45.64, 52.46)

μ : population mean of PS_CSK wins only

After conducting a 1-Sample t-Confidence Interval analysis using both Minitab and R. The dataset was filtered to include only matches where CSK won. The Powerplay scores from these matches were used to calculate a 95% confidence interval for the mean.

The result shows 95% confidence that the true average Powerplay score in matches that CSK won lies between 45.64 and 52.46 runs.

This implies that in order to maximize their winning chances, CSK should aim to score at least around 46–52 runs in the Powerplay overs. This range can be used as a benchmark

when strategizing batting approaches for future matches.

b. Estimate the expected first innings score at different IPL venues using 95% Confidence Intervals, helping CSK strategize venue-specific scoring targets.

Variable	Venue	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
First_Innings_Score		0	924	*	*	*	*	*	*	*	*
	ACA-VDCA Cricket Stadium, Visakhapatnam	1	0	191	*	*	191	191	191	191	191
	Arun Jaitley Stadium, Delhi	3	0	204	16.5630	28.6880	171	171	218	223	223
	Brabourne Stadium, Mumbai	3	0	180	17.3205	30	150	150	180	210	210
	Dr DY Patil Sports Academy, Mumbai	4	0	183.25	16.6802	33.3604	154	154.25	181.5	214	216
	Dubai International Cricket Stadium	12	0	163.333	5.05025	17.4946	134	147.75	168	174.25	192
	Eden Gardens, Kolkata	1	0	235	*	*	235	235	235	235	235
	Himachal Pradesh Cricket Association Stadium, Dharamsala	1	0	167	*	*	167	167	167	167	167
	M Chinnaswamy Stadium, Bengaluru	1	0	226	*	*	226	226	226	226	226
	M. A. Chidambaram Stadium, Chennai	1	0	141	*	*	141	141	141	141	141
	M. Chinnaswamy Stadium, Bengaluru	1	0	218	*	*	218	218	218	218	218
	MA Chidambaram Stadium, Chepauk, Chennai	14	0	174.857	7.97376	29.8351	134	142.75	172.5	207	217
	Maharashtra Cricket Association Stadium, Pune	3	0	181.333	10.3976	18.0093	169	169	173	202	202
	Narendra Modi Stadium, Ahmedabad	3	0	207.667	15.6241	27.0617	178	178	214	231	231
	Rajiv Gandhi International Stadium, Hyderabad	1	0	165	*	*	165	165	165	165	165
	Sawai Mansingh Stadium, Jaipur	1	0	202	*	*	202	202	202	202	202
	Sharjah Cricket Stadium	5	0	159.8	17.7550	39.7014	114	124	156	197.5	216
	Sheikh Zayed Stadium	4	0	151.75	9.37528	18.7506	125	132	157.5	165.75	167
	Vajpayee Cricket Stadium, Lucknow	2	0	150.5	25.5	36.0624	125	125	150.5	176	176
	Wankhede Stadium, Mumbai	11	0	164	12.5271	41.5476	97	131	187	191	220
	Zayed Cricket Stadium, Abu Dhabi	2	0	180	9	12.7279	171	171	180	189	189

Analysis of the Chennai Super Kings' (CSK) first innings scores at various regularly visited IPL locations using a 95% Confidence Interval (CI). This aids in estimating the range that CSK's average score should fall into 95% of the time.

The expected range for Chennai's MA Chidambaram Stadium is between 160–190, indicating steady play and home comfort. Greater variability is indicated by the broader range (~139–189) at Wankhede Stadium in Mumbai. The greatest mean (~208) and widest confidence interval (~177–238) are found in Narendra Modi Stadium in Ahmedabad, indicating explosive potential. These revelations aid in the comprehension of anticipated scoring patterns and venue dynamics in relation to CSK's batting performances.

2. Confidence Interval of Difference of Mean

CSK First Innings Score (Wins vs. Losses)

Using data from the last five IPL seasons, performed a two-sample t-test to investigate whether Chennai Super Kings (CSK) typically score considerably more runs in the first innings when they win as opposed to when they lose. First innings scores from games that CSK won (Group 1) and those that CSK lost (Group 2) were used to separate the sample.

Descriptive Statistics: First_Innings_Score					Estimation for Difference		Test		
Winner Team (Own/Opp)	N	Mean	StDev	SE Mean	Difference	95% CI for Difference	T-Value	DF	P-Value
1	11	171.2	22.3	6.7	5.4	(-16.3, 27.0)	0.52	19	0.610
2	11	165.8	26.0	7.8					

- The **mean first innings score** in winning matches was **171.2** with a standard deviation of **22.3**.
- The **mean score** in losing matches was **165.8** with a standard deviation of **26.0**.
- The estimated **difference in means** was **5.4 runs**, and the **95% confidence interval** for this difference ranged from **-16.3 to 27.0**.
- The **p-value** obtained was **0.610**, which is well above the significance threshold of 0.05.

The null hypothesis cannot be rejected since the p-value is not statistically significant and the confidence interval contains 0. This implies that CSK's first innings score does not differ statistically significantly between their winning and losing games. As a result, it doesn't seem like CSK's performance in the first innings alone will determine the outcome of the match.

3. Confidence Interval for Proportions

To estimate the proportion of matches CSK wins when they win the toss, and compute a 95% confidence interval for this proportion.

Descriptive Statistics

N	Event	Sample p	95% CI for p
33	19	0.575758	(0.401217, 0.732501)

Over the previous five seasons, we aimed to calculate the percentage of games that CSK won after winning the toss. CSK won 19 of the 33 of these games.

With a 95% confidence level, the sample proportion ($\hat{p}$) was 0.576, which indicates that CSK won the toss roughly 57.6% of the time.

The true population proportion's 95% confidence interval is between 0.401 and 0.733, meaning that we have a 95% chance that the actual percentage of CSK that wins after winning the toss falls between 40.1% and 73.3%.

Hypothesis Testing:

In order to determine whether Chennai Super Kings (CSK) exhibit statistically significant dominance under either approach, the project's premise is based on analyzing the team's victory percentages in two distinct match scenarios: first and second. In order to evaluate strategic dominance, the two situations are also compared.

It was assessed how the win proportions from the two samples differed. If the null hypothesis, which would suggest statistically significant performance advantage in a particular case, can be rejected, the hypothesis is deemed successful.

A) When Batting First, CSK Wins : The Chennai Super Kings' victory percentage when they bat first in IPL games from 2020 to 2024 is assessed by this hypothesis test.

- Null Hypothesis (H_0): When CSK bats first, the victory percentage is less than or equal to 0.5.
- Alternative Hypothesis (H_1): When CSK bats first, the victory percentage is higher than 0.5
- Interest-related parameter: p

0.5 is the null hypothesis, and $H_0 = p$

Hypothesis Alternative: $H_1 = p > 0.5$

CSK won 18 of the 27 games in which they most usually batted first, resulting in a sample proportion of 0.6667.

A p-value of 0.0272, which is below the significance level of 0.05, was obtained from the test. Therefore, we accept the alternative hypothesis and reject the null hypothesis. This implies compelling statistical proof that CSK's winning percentage when batting first is far higher than 50%. The notion that CSK performs better when defending a target is supported by this.

B) CSK Wins when Batting Second: Whether CSK maintains a statistically significant win rate when chasing (batting second) is ascertained by this test.

- The victory percentage when CSK bats second is less than or equal to 0.5, which is the **null hypothesis (H_0)**.
- Alternative Hypothesis (H_1): When CSK bats second, the victory percentage is higher than 0.5
- Interest-related parameter: p

0.5 is the null hypothesis, and $H_0 = p$

Hypothesis Alternative: $H_1 = p > 0.5$

CSK won 21 of the 33 games in which they batted second, yielding a sample proportion of 0.636.

A p-value of 0.0333, which is likewise less than 0.05, was obtained from the test.

Consequently, the alternative hypothesis is accepted and the null hypothesis is rejected. This demonstrates that when chasing targets, CSK also wins noticeably more than half of the games. It shows the team's adaptability in different match strategies and consistency under duress.

C) Comparison : Batting First vs Batting Second: This last test determines whether either method offers a statistically significant advantage by comparing the win proportions between the two situations.

The **null hypothesis (H_0)** states that the victory percentage for CSK when they bat first is the same as the win percentage when they bat second. The **alternative hypothesis (H_1)** states that The percentages of wins are not equal.

- Interest-related parameter: $p_1 - p_2$
- Nothing Assumption: $H_0 = p_1 - p_2 = 0$
- Hypothesis Alternative: $H_1 = p_1 - p_2 \neq 0$

First-batting: 18 victories in 27 games (66.67%)

Batting second: 21 victories (63.63%) in 33 games.

The p-value for the two-sample proportion test was 0.795, which is significantly higher than 0.05.

As a result, we are unable to reject the null hypothesis and it is accepted.

This suggests that CSK's win percentages while batting first and second do not differ statistically significantly. As a result, CSK exhibits flexibility and general match awareness by performing consistently in all match situations.

ANOVA

In order to determine if there are any notable differences in CSK's Powerplay scores (overs 0–6) amongst IPL venues, a one-way Analysis of Variance (ANOVA) was performed in Minitab. Venue, which included 20 different stadiums, was the categorical variable evaluated, while Powerplay_Scores was the numerical response variable.

The following were the definitions of the hypotheses examined in the analysis:

The hypotheses tested in the analysis were defined as follows:

- Null Hypothesis (H_0): The average Powerplay scores are identical across all venues.

- Alternative Hypothesis (H_1): At least one venue has a different mean Powerplay score.

An F-statistic of 2.30 and a p-value of 0.009 were produced by the ANOVA output. The null hypothesis is rejected when the p-value is less than the conventional significance level of 0.05. This offers enough proof to draw the conclusion that CSK's Power Play scoring is greatly impacted by the venue. These discrepancies could be caused by stadium-specific features, environmental factors, or pitch behavior.

Tukey's test results are shown above. Different letters indicate statistically different venues. Disparities in performance are confirmed by the fact that Brabourne and Jaipur are isolated from locations like Lucknow.

Grouping Information Using the Tukey Method and 95% Confidence

Venue	N	Mean	Grouping	
Brabourne Stadium, Mumbai	3	73.333	A	
Sawai Mansingh Stadium, Jaipur	1	64.00	A	B
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium, Visakhapatnam	1	62.00	A	B
Eden Gardens, Kolkata	1	59.00	A	B
M. Chinnaswamy Stadium, Bengaluru	1	58.00	A	B
M. A. Chidambaram Stadium, Chennai	1	56.00	A	B
MA Chidambaram Stadium, Chepauk, Chennai	14	53.07	A	B
M Chinnaswamy Stadium, Bengaluru	1	53.00	A	B
Narendra Modi Stadium, Ahmedabad	3	52.00	A	B
Himachal Pradesh Cricket Association Stadium, Dharamsala	1	51.00	A	B
Sheikh Zayed Stadium	4	49.75	A	B
Rajiv Gandhi International Stadium, Uppal, Hyderabad	1	48.00	A	B
Zayed Cricket Stadium, Abu Dhabi	2	47.00	A	B
Arun Jaitley Stadium, Delhi	3	46.67	A	B
Maharashtra Cricket Association Stadium, Pune	3	45.33	A	B
Dr DY Patil Sports Academy, Mumbai	4	43.75		B
Wankhede Stadium, Mumbai	11	42.73		B
Sharjah Cricket Stadium	5	42.60		B
Dubai International Cricket Stadium	12	41.75		B
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cricket Stadium, Lucknow	2	41.0	A	B

Means that do not share a letter are significantly different.

Fig 12: Tukey test

Grouping Information Using the Tukey Method and 95% Confidence

Venue	N	Mean	Grouping	
Brabourne Stadium, Mumbai	3	73.333	A	
Sawai Mansingh Stadium, Jaipur	1	64.00	A	B
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium, Visakhapatnam	1	62.00	A	B
Eden Gardens, Kolkata	1	59.00	A	B
M. Chinnaswamy Stadium, Bengaluru	1	58.00	A	B
M. A. Chidambaram Stadium, Chennai	1	56.00	A	B
MA Chidambaram Stadium, Chepauk, Chennai	14	53.07	A	B
M Chinnaswamy Stadium, Bengaluru	1	53.00	A	B
Narendra Modi Stadium, Ahmedabad	3	52.00	A	B
Himachal Pradesh Cricket Association Stadium, Dharamsala	1	51.00	A	B
Sheikh Zayed Stadium	4	49.75	A	B
Rajiv Gandhi International Stadium, Uppal, Hyderabad	1	48.00	A	B
Zayed Cricket Stadium, Abu Dhabi	2	47.00	A	B
Arun Jaitley Stadium, Delhi	3	46.67	A	B
Maharashtra Cricket Association Stadium, Pune	3	45.33	A	B
Dr DY Patil Sports Academy, Mumbai	4	43.75		B
Wankhede Stadium, Mumbai	11	42.73		B
Sharjah Cricket Stadium	5	42.60		B
Dubai International Cricket Stadium	12	41.75		B
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cricket Stadium, Lucknow	2	41.0	A	B

Means that do not share a letter are significantly different.

Fig 13: 95% Confidence and Tukey grouping

Confidence intervals for mean Powerplay scores by venue are shown above. Broader intervals in Sharjah suggest more fluctuation, while small intervals are seen at consistent locations such as Chepauk and Brabourne.

Overall, the results show that CSK's Powerplay production is greatly influenced by the match venue. Early scoring momentum is facilitated by high-performing venues, underscoring the significance of taking venue features into account when developing match strategy and performance analysis.

Results:

1. Sampling Distribution:

- a) Means-Powerplay Scores: CSK's Powerplay average over 1000 samples ($n = 30$) has a bell-shaped distribution with a centre of 48.
- b) Variance-Death Over Scores: CSK's spectacular but erratic finishes are reflected in this right-skewed distribution, which peaks between 175 and 200.
- c) Wins when Batting First: The majority of proportions are less than 0.25, which suggests a lower victory rate when establishing goals.
- d) Wins while Batting Second: The proportions show that CSK is more successful when pursuing goals, clustering between 0.35 and 0.40.

2. Confidence Intervals:

- a) Powerplay Scores in Wins: For a better chance of winning, CSK should aim for 45.64 to 52.46 runs in Powerplay.
- b) Score Ranges for First Innings by Venue:
 - i) Chennai's MA Chidambaram Stadium: 160–190
 - ii) Mumbai's Wankhede Stadium: 139–189
 - iii) Ahmedabad's Narendra Modi Stadium: 177–238
- c) First Innings Score Difference (Wins vs. Losses): 5.4 runs; $p\text{-value} = 0.610 \Rightarrow$ Not statistically significant.
- d) Win percentage following a toss: 57.6%; $CI = 0.401, 0.733$

3. Hypotheses Testing:

- When CSK bats first, they win: 66.7% of the total, $p = 0.0272$, $\rightarrow$ Significant.
- When CSK bats second, they win: $p = 0.0333 \Rightarrow$ Significant, proportion = 63.6%.
- Comparing First and Second Batting: $p = 0.795 \rightarrow$ Not much of a difference.

4. ANOVA (One Way):

- Powerplay scores vary significantly by venue ($p = 0.009$).
- Brabourne and Jaipur have statistically greater Powerplay scores than locations like Lucknow, according to Tukey HSD.
- The venue affects CSK's early match momentum, highlighting the significance of the pitch and surrounding circumstances.

Conclusions:

- **Batting Second Advantage:** When pursuing objectives, CSK has a statistically significant greater win rate.
- **Powerplay Benchmark:** Winning matches are highly correlated with scoring 46–52 runs in Powerplay.
- **Venue Matters:** While locales like Lucknow may impair performance, places like Brabourne, Jaipur, and Ahmedabad provide better scoring circumstances.
- **Despite winning 57.6% of toss-win games,** CSK's statistical significance is insufficient to support their use as a reliable performance driver.
- **Score in the First Innings Not a Sole Factor:** Wins and defeats did not significantly differ in first innings scores, indicating that other stages or match circumstances are more important.

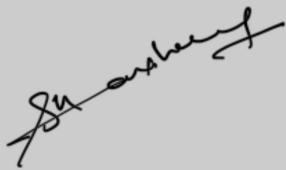
Limitations:

- **Absence of Player-Level Analysis:** Individual performance measures (such as bowling economy and strike rate) are not as detailed.
- **No External Conditions Were Taken Into Account:** The pitch report, crowd influence, and weather were not taken into account.
- **Restricted Seasons:** Only five seasons (2020–2024) are included in the dataset, which might not adequately represent long-term trends.
- **Focus on Binary Outcomes:** Win/Loss as the sole outcome statistic disregards close finishes or match competition.
- **Static Match Importance:** Gives every match the same weight, disregarding playoffs, championships, or games that must be won.

Proposed Next Steps and Future Work:

1. **Include Metrics at the Player Level:** To have a deeper understanding, include variables such as bowler economy, strike rates, top batsman scores, and dot ball percentages.
2. **Modelling Phase-Wise Strategies:** To more dynamically examine how each phase (Powerplay, Middle, and Death) affects results, use regression or machine learning.
3. **Integration of Weather and Pitch Data:** Incorporate temperature, dew conditions, and pitch type (pace vs. slow/spin-friendly) to improve venue-based tactics.
4. **Simulations Based on Scenarios:** Use bootstrap sampling to simulate match results under fictitious conditions (such as chasing 180 at Eden Gardens).
5. **Extension to Additional Teams for Comparison:** To put CSK's strengths and shortcomings in perspective, compare their statistics to those of clubs like MI, GT, or RCB.

Collaboration

Team Member	Contributions	Signature*
Sai Ashok Madugula	Conducted Confidence Interval Analysis using R & Minitab and supported overall Project Report drafting. CONTRIBUTION-25%	
Tanisha Kishan	Prepared Data Visualization charts for overall performance of CSK , Conducted Anova,Addressing the Feedback and structuring the final report. CONTRIBUTION-25%	
Shamsheer Hussain Syed	Contributed in preparing visualization charts. Worked on Results/Conclusions and Limitations for the Final Project report. CONTRIBUTION-25%	
Dheeraj Kumar Kusuma	Worked on the major part of the Statistical Analysis. (1. Sampling Distributions of Means, Variances and proportions, 2. Hypotheses Testing) on R, Proposed Next steps and Future Works. Worked on addressing the feedback (Data Visualization). CONTRIBUTION-25%	

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